

Policy considerations for effective EV-based evacuations

Improving Preparedness of Communities for Evacuations using ZEVs

2nd Stakeholder Workshop

December 5, 2024

Motivation and Aims



Inform our modeling and analysis activities



Assess and compare policy and planning strategies



Identify preferences and feasibility of different policies/planning

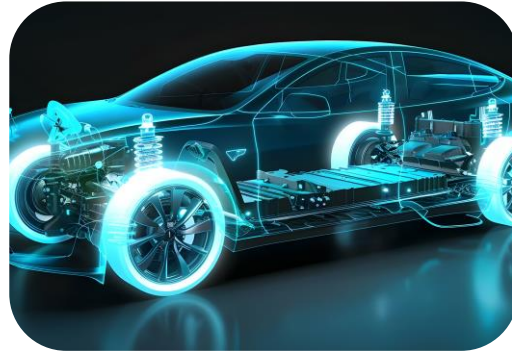


Provide guidance and recommendations for policies and practices that support positive ZEV evacuation outcomes

Policy and planning options span multiple sectors and scales



Charging
Infrastructure



Vehicles



Transportation
Infrastructure



Energy
Infrastructure

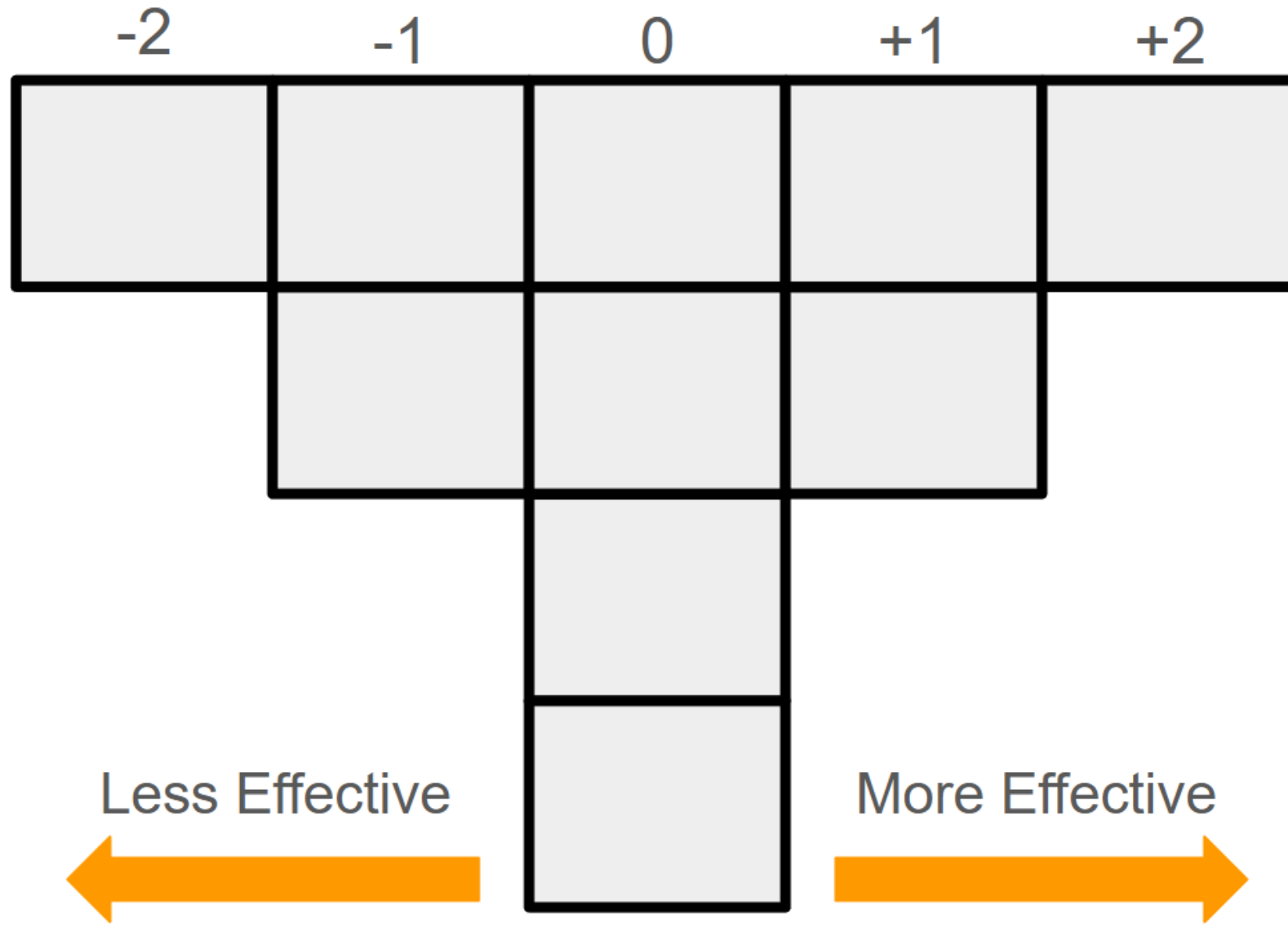
Overview of strategies for consideration

Strategy #	Strategy	Description
1	Backup energy – fossil fuels	Require public charging stations to have on-site backup energy to maintain operability during power outages. The backup power would come from diesel generators (or similar)
2	Backup energy - microgrid	Require public charging stations to have on-site backup energy to maintain operability during power outages. The backup power would come from microgrids consisting of some combination of solar/wind energy and battery energy storage
3	Free/open charging	During emergency declarations, public charging stations would be “unlocked” so that people can essentially drive up and immediately start charging without having to pay or interface with any kind of app or point of service system. Charging companies could be reimbursed by federal/state governments
4	Mobile chargers at origin	Dispatch mobile chargers (of any type) to evacuation zones prior to the event to increase the initial state of charge and reduce the need for charging along the evacuation route
5	Mobile chargers to supplement existing charging locations	Dispatch mobile charges (of any type) to sites with existing stationery charging infrastructure. The mobile chargers would act as supplemental resources to the existing charging infrastructure

Overview of strategies for consideration (continued)

Strategy #	Strategy	Description
6	Mobile chargers at designated locations along evacuation route	Dispatch mobile chargers (of any type) to designated locations (e.g., rest stops, shopping centers, etc.) along evacuation routes that have minimal existing charging infrastructure
7	Minimum “charge reserve” required on all EVs	On all EVs, require a certain % of battery charge to be kept in reserve that can be unlocked during an emergency declaration
8	Bussing/shuttling people out of evacuation zones	Use buses or shuttles (or any type) to facilitate the evacuation of people rather than relying on personal vehicles. This could reduce congestion on the road, as well as demand at downstream charging stations
9	Staggered departures from evacuation zones	Specify time periods when different locations can evacuate. This could reduce congestion on the road, as well as demand at downstream charging stations
10	Minimum “spinning reserves” required in the power grid	Require utility companies and electricity grid operators to keep a certain amount of electricity generation capacity on “standby”. This can support a rapid uptick in charging demand prior to and during an evacuation. It also provides some “backup” capacity in reduced generation from solar/wind

Intro to Q Sort Methodology



We'd love to have your input and perspective...

<https://tinyurl.com/yc7cr2ef>

Thank you!!!

- Please don't hesitate to reach out if you have questions, comments, suggestions, or ideas

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